

WEBB CITY JOPLIN, MO, USA

WMO: 723495

Lat: 37.152N

Lon: 94.495W

Elev: 296

StdP: 97.82

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13987

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.9	-10.2	-18.3	0.8	-11.5	-15.7	1.0	-8.2	12.5	12.0	11.6	11.0	3.7	10	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	36.3	23.9	34.6	24.2	33.1	24.0	26.1	32.5	25.5	31.9	24.9	31.1	4.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.9	29.9	23.6	19.1	29.2	23.0	18.3	28.5	82.4	32.7	79.7	32.0	77.1	31.1	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	9.9	8.7	-17.0	38.0	2.9	1.9	-19.1	39.4	-20.7	40.5	-22.3	41.5	-24.4	42.9		
(5)			-17.5	27.1	2.7	1.0	-19.4	27.9	-21.0	28.5	-22.6	29.1	-24.6	29.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.8	1.9	4.3	9.3	14.7	19.4	24.4	26.8	26.4	21.9	15.5	9.2	3.6
	DBStd	10.08	6.66	6.59	6.31	5.18	4.35	3.15	2.98	3.15	4.25	5.17	5.69	6.00
	HDD10.0	852	262	181	89	16	1	0	0	0	0	12	81	211
	HDD18.3	2228	510	394	284	132	41	2	0	1	16	117	276	456
	CDD10.0	2622	11	21	68	157	294	433	521	508	356	184	58	13
	CDD18.3	954	0	1	5	22	75	184	262	250	122	31	3	0
	CDH23.3	9739	0	5	38	178	621	1826	2968	2721	1138	229	17	0
	CDH26.7	4214	0	1	4	32	162	737	1437	1341	447	52	1	0
Wind	WSAvg	4.1	4.4	4.5	4.8	5.0	4.2	3.9	3.4	3.3	3.3	3.9	4.4	4.2
Precipitation	PrecAvg	1108	44	56	88	104	127	143	82	97	128	92	88	74
	PrecMax	1657	105	199	230	264	327	309	316	217	485	232	313	195
	PrecMin	466	0	6	21	3	37	21	4	1	25	1	1	12
	PrecStd	252	30	41	49	56	67	65	66	59	101	58	70	48
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.9	23.7	26.8	29.6	31.8	35.6	38.3	38.7	36.1	30.6	25.6	21.0
		MCWB	14.2	15.6	16.8	18.6	22.4	24.2	23.4	23.9	22.7	21.7	17.3	15.6
	2%	DB	17.7	20.2	24.1	27.3	30.1	33.5	36.2	36.5	32.8	28.0	22.5	18.2
		MCWB	12.1	13.4	15.9	18.0	21.9	24.2	24.1	23.9	22.8	19.5	15.8	13.4
	5%	DB	15.3	17.7	21.7	25.4	28.6	32.1	34.6	34.7	31.2	26.0	20.5	15.5
		MCWB	10.4	12.3	14.4	17.0	21.3	23.8	24.7	24.0	22.6	18.3	15.2	11.3
	10%	DB	12.1	15.1	19.5	23.4	27.0	30.8	33.0	33.0	29.3	23.8	18.2	12.9
		MCWB	8.0	9.8	13.5	16.4	20.6	23.4	24.6	23.6	21.5	17.9	13.4	8.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.2	17.3	18.9	20.8	24.1	26.3	27.0	26.7	25.2	22.9	19.2	16.9
		MCDB	19.2	20.7	24.0	26.6	29.4	32.8	33.2	33.1	31.3	28.2	22.8	20.1
	2%	WB	13.8	15.4	17.2	19.7	23.2	25.4	26.3	25.7	24.1	21.6	17.4	14.7
		MCDB	16.7	19.2	22.1	24.8	28.3	31.4	32.7	32.6	30.7	25.8	21.1	17.5
	5%	WB	10.7	13.0	15.8	18.7	22.3	24.6	25.7	25.1	23.4	20.3	15.8	12.1
		MCDB	14.3	16.3	20.5	23.5	27.2	30.5	32.0	32.0	29.5	23.9	19.2	14.6
	10%	WB	7.9	10.2	14.1	17.5	21.2	23.9	25.2	24.4	22.5	18.9	14.2	9.6
		MCDB	11.8	14.8	18.9	22.1	25.7	29.4	31.3	30.9	27.8	22.8	17.6	12.4
Mean Daily Temperature Range	5% DB	MDBR	10.5	11.0	11.5	11.6	10.8	10.6	10.7	11.4	11.7	11.7	11.1	9.9
		MCDBR	13.8	14.6	14.1	13.3	11.6	11.5	12.2	13.0	12.5	12.9	12.9	12.9
		MCWBR	9.0	9.6	8.1	7.2	5.8	4.8	4.0	4.2	5.0	6.3	8.1	8.9
	5% WB	MCDBR	12.3	13.0	11.6	10.8	10.5	10.6	10.5	10.9	10.9	10.0	11.7	11.3
		MCWBR	9.4	10.0	7.8	6.9	5.8	5.1	4.5	4.5	4.8	5.7	8.6	9.1
Clear-Sky Solar Irradiance	taub	0.304	0.313	0.346	0.380	0.423	0.423	0.440	0.443	0.387	0.345	0.311	0.291	
	taud	2.449	2.437	2.388	2.313	2.250	2.280	2.277	2.246	2.369	2.464	2.502	2.507	
	Ebn at Noon	886	923	918	900	862	859	841	831	863	874	873	881	
	Edh at Noon	84	96	111	127	137	133	133	134	112	93	79	74	
All-Sky Solar Radiation	RadAvg	2.40	2.98	3.97	5.04	5.65	6.50	6.46	5.80	4.92	3.60	2.57	2.04	
	RadStd	0.21	0.45	0.25	0.33	0.47	0.53	0.40	0.38	0.36	0.43	0.28	0.24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page